Menna:

- Identified data quality issues including missing values and inconsistent column naming
- Removed columns with much nulls (Attractions, FaxNumber, Map, HotelWebsiteUrl, countyCode, cityCode, HotelCode)
- removing leading spaces from columns headers
- Selected most important columns for analysis: countyName, cityName, HotelName, HotelRating, Address, Description, HotelFacilities, PhoneNumber, PinCode
- Dropped rows with missing values, reducing dataset from 1,010,033 to 651,079 rows
- Validated the cleaned dataset schema using Pandera library
- Generated Sweetviz profiling report to document data information
- Saved cleaned dataset to “cleaned_hotels.csv”
- Hash the `PhoneNumber` column using SHA-256
- Define functions: `is_prime()`, `get_primes()`, `gcd()`, `mod_inverse()`
- Define RSA functions: `generate_rsa_keys()`, `rsa_encrypt()`, `rsa_decrypt()`
- Encrypt `HotelName` column and store in `Encrypted_HotelName`
- Decrypt back to verify and store in `Decrypted_HotelName`
- Create a separate DataFrame `encrypt_decrypt_hotelname` showing encryption/decryption results
- Drop original `HotelName` and `Decrypted_HotelName` columns from main DataFrame
- Extract `real_PhoneNumber` into a separate DataFrame
- Drop the original `PhoneNumber` column
- Create HLL (HyperLogLog) sketch to estimate unique city count
- Create Kll sketch for estimation (min, max, median of indices)

Bonsus:

- Factorize RSA modulus `n` to recover `p` and `q`
- Recompute `phi` and `d` from factored primes
- Decrypt all encrypted hotel names using the recovered private key

Rozina:

- Created bar chart showing top 10 countries by hotel count
- Created bar chart showing top 10 cities by hotel count
- Built average rating by country chart to identify highest-rated destinations
- Converted text-based HotelRating values to numeric format for proper analysis
- Mapped rating values: FiveStar→5, FourStar→4, ThreeStar→3, TwoStar→2, OneStar→1
- Handled "All" data value by mapping to 5-star rating
- Generated horizontal bar charts for better readability of country and hotel rankings
- Added summary statistics displays showing total hotels, average rating, and unique counts
- Create a `users_df` with user IDs, names, and IP addresses
- Define `check_access()` function using simple RBAC logic (allowed users, actions, IP filtering)
- Test access for user "Menna" – Access granted
- Implement full ABAC using `py_abac`

- Create audit log DataFrame with timestamps, users, resources, actions, IPs, authorization status, HIPAA flags
- Flag HIPAA violations (unauthorized access attempts)
- Print compliance summary table (CCPA/HIPAA, libraries used, steps, status)

Bonus:

- drop non-feature columns
- Map `HotelRating` from text to numeric values (5,4,3,2,1)
- Map ‘All Star’ to 5 stars
- Encode categorical columns (`countyName`, `cityName`) using `LabelEncoder`
- Remove `PhoneNumber` column ‘
- Remove `PinCode` column
- Define features (countyName_Encoded, cityName_Encoded) and target (HotelRating)